

B.Sc. (Hons.) Semester IV Examination 2023-24
STATISTICS

Paper No. STB-402: Practical based on STB-401 (SET-II)

Time: Three hours

Max. Marks: 70

Note: Attempt any three questions in all. All questions carry equal marks. Use of Excel software is permitted for computation.

1. A population consists of six units with respective values 31, 30, 42, 36, 28, and 38. List all possible random samples of size 2 using Simple Random Sampling With Replacement (SRSWR). Also, show empirically that:
 - (a) The sample mean $\bar{y}$ is an unbiased estimator of the population mean $\bar{Y}$.
 - (b) The sample mean square s^2 is an unbiased estimator of the population variance σ^2 .
 - (c) $V(\bar{y}) = \frac{N-1}{Nn} S^2$, verifying with structural parameters.
2. A sample survey is to be undertaken to ascertain the mean annual income of farms in a certain area. The farms are stratified according to their principal products. A census conducted several years earlier gave the following baseline information:

Type of farm	No. of farms (N_i)	Mean annual income ($\bar{Y}_{Ni}$)	Standard deviation (S_i)
Sheep	161	10,946	2,236
Wheat	195	6,402	2,614
Dairying	274	2,228	606
Others	382	1,458	230

A sample of 90 farms is to be selected from this population. Compute the required sample sizes for each stratum under:

- (a) Proportional allocation
- (b) Neyman (Optimum) allocation

Also, calculate and compare the sampling variance of the estimator of the population mean under both allocation schemes.

3. An investigator was interested in estimating the average amount of real estate farm loans (in \$000) during 1997 in the United States. He took an SRSWOR sample of eight states from the population of 50 states and collected the following information:

State	CA	GA	LA	MS	NM	PA	TX	VT
Non-real estate loans (X)	3,928.73	540.70	405.80	549.55	274.04	298.35	3,520.36	19.36
Real estate loans (Y)	1,343.46	939.46	282.57	627.01	140.58	756.17	1,248.76	57.75

The true population average amount of non-real estate farm loans ($\bar{X}$) for the year 1997 is known to be 878.16.

- (a) Apply the ratio method of estimation to estimate the population average amount of real estate farm loans ($\bar{Y}$).
- (b) Compute the estimates of the Mean Square Error (MSE) of the ratio estimator and the variance of the mean per unit estimator under SRSWOR.

- (c) Evaluate the relative efficiency of the ratio estimator with respect to the mean per unit estimator.
4. The table below gives the field layout and yields for five varieties of wheat obtained in an agricultural experiment consisting of four randomized blocks:

Variety	Block I	Block II	Block III	Block IV
A	27.8	27.4	28.5	38.5
B	36.4	28.8	31.2	39.4
C	24.8	27.6	34.6	36.8
D	16.4	15.3	14.4	19.4
E	18.6	17.4	17.8	15.6

Perform a complete two-way analysis of variance (ANOVA) on this dataset at a 5% level of significance, test the significance of block differences and variety effects, and present your findings.

External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>